Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Live Oak Dairy G

Physical address of dairy:

Number and Street	City	County	Zip Code
730 Bystrum RD	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0057-0025-0012-0000 X022-X038-X009-XXXX X045-X030-X018-XXXX

B. OPERATORS

Nyman, Brad

Operator name: Nyman, Brad

Telephone no.: (209) 495-8821 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 1331	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Melo, Luis C

Legal owner name: Melo, Luis C

Telephone no.: (209) 604-3268 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
730 Bystrum RD	Turlock	CA	95380

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	150	100	0	0	0
Number under roof	1,200	0	0	0	0	0
Maximum number	1,230	150	100	0	0	0
Average number	1,200	150	100	0	0	0
Avg live weight (lbs)	950	1,000	700	0		

Predominant milk cow breed: Jersey

Average milk production: 50 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 30,815.31 *tons per reporting period*

Total nitrogen from manure: 402,494.08 *lbs per reporting period*

After ammonia losses (30% loss applied): 281,745.86 *lbs per reporting period*

Total phosphorus from manure: 66,581.42 *lbs per reporting period*

Total potassium from manure: 214,043.07 *lbs per reporting period*

Total salt from manure: 599,512.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 11,115,000 *gallons*

Total nitrogen generated: 72,276.17 *lbs*

Total phosphorus generated: 6,786.70 *lbs*

Total potassium generated: 51,492.11 *lbs*

Total salt generated: 571,829.44 *lbs*

	11,115,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	11,115,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Lateral 4	Surface water
TID Lateral 6	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Bystrum 1	19	19	2	process wastewater	X022-X038-X008-XXXX
Bystrum 2	35	35	2	process wastewater	X022-X038-X009-XXXX
Bystrum 3	21	21	2	process wastewater	X022-X038-X009-XXXX
Bystrum 4	29	29	2	process wastewater	X022-X039-X011-XXXX
Bystrum 5	12	12	2	process wastewater	X022-X038-X009-XXXX
Central 1	57	57	2	manure	X057-X025-X006-XXXX X057-X025-X007-XXXX
Central 2	35	35	2	manure	X057-X025-X007-XXXX
Central 3	78	78	2	manure	X045-X030-X018-XXXX X057-X025-X007-XXXX
Central 4	69	69	2	manure	X045-X030-X018-XXXX X057-X025-X006-XXXX X057-X025-X007-XXXX
Central 5	20	20	2	manure	X057-X025-X006-XXXX
Central 6	66	66	2	manure	X057-X025-X006-XXXX
Totals for areas that were used for application	441	441	22		
Totals for areas that were not used for application					
Land application area totals	441	441	22		

B. CROPS AND HARVESTS

Bystrum 1

Field name: Bystrum 1

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	364.00 *ton*	As-is		59.8	7,120.00	1,850.00	23,300.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	19.16	272.81	70.88	892.76	2,094.80

Bystrum 1

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	642.00 *ton*	As-is		63.6	3,440.00	730.00	6,190.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	33.79	232.47	49.33	418.31	1,106.94

Bystrum 2

Field name: Bystrum 2

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	812.00 *ton*	As-is		69.8	5,290.00	1,630.00	23,300.00		13.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	23.20	245.46	75.63	1,081.12	1,863.70

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	1,056.00 *ton*	As-is		60.4	4,430.00	910.00	7,130.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.17	267.32	54.91	430.24	1,051.41

Bystrum 3

Field name: Bystrum 3

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	406.00 *ton*	As-is		64.2	8,130.00	2,040.00	26,100.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	19.33	314.36	78.88	1,009.20	1,868.76

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	703.00 *ton*	As-is		58.0	4,510.00	1,090.00	8,820.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	33.48	301.96	72.98	590.52	1,434.12

Bystrum 4

Field name: Bystrum 4

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	657.00 *ton*	As-is		71.6	5,680.00	1,330.00	21,000.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	22.66	257.36	60.26	951.52	1,698.59

Bystrum 4

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	896.00 *ton*	As-is		62.4	4,410.00	980.00	8,270.00		5.11

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.90	272.51	60.56	511.03	1,187.27

Bystrum 5

Field name: Bystrum 5

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	221.00 *ton*	As-is		71.2	7,290.00	1,670.00	21,000.00		13.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.42	268.52	61.51	773.50	1,410.86

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	400.50 *ton*	As-is		66.0	3,970.00	850.00	7,140.00		4.59

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	33.38	265.00	56.74	476.60	1,041.70

Central 1

Field name: Central 1

12/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 57 Plant date: 12/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/10/2023	665.00 *ton*	As-is		60.6	7,170.00	2,170.00	27,600.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	11.67	167.30	50.63	644.00	965.30

07/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 07/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	702.00 *ton*	As-is		64.2	6,330.00	680.00	13,600.00		6.79

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	12.32	155.92	16.75	334.99	598.75

Central 2

Field name: Central 2

12/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 12/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/10/2023	487.00 *ton*	As-is		66.0	6,900.00	1,840.00	24,500.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	13.91	192.02	51.20	681.80	1,192.18

Central 2

07/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 07/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	485.00 *ton*	As-is		59.8	6,240.00	600.00	6,430.00		5.47

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	13.86	172.94	16.63	178.20	609.42

Central 3

Field name: Central 3

12/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 78 Plant date: 12/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/10/2023	789.00 *ton*	As-is		54.8	8,450.00	2,350.00	31,200.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	10.12	170.95	47.54	631.20	1,234.48

07/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 78 Plant date: 07/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	1,155.00 *ton*	As-is		61.8	5,500.00	570.00	4,580.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	14.81	162.88	16.88	135.64	509.09

Central 4

Field name: Central 4

12/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 69 Plant date: 12/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/10/2023	809.00 *ton*	As-is		63.4	6,700.00	1,790.00	24,500.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	11.72	157.11	41.97	574.51	1,029.89

07/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 69 Plant date: 07/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/14/2023	1,145.00 *ton*	As-is		60.8	4,350.00	510.00	3,920.00		3.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	16.59	144.37	16.93	130.10	390.30

Central 5

Field name: Central 5

12/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 12/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/11/2023	221.00 *ton*	As-is		67.4	7,070.00	1,990.00	26,100.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	11.05	156.25	43.98	576.81	778.10

Central 5

07/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 07/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	391.00 *ton*	As-is		65.0	3,940.00	630.00	4,550.00		4.61

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	19.55	154.05	24.63	177.91	630.88

Central 6

Field name: Central 6

12/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 66 Plant date: 12/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/11/2023	703.00 *ton*	As-is		67.0	8,280.00	1,680.00	20,100.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	10.65	176.39	35.79	428.19	927.96

07/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 66 Plant date: 07/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	1,072.00 *ton*	As-is		61.6	4,540.00	580.00	4,610.00		4.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	16.24	147.48	18.84	149.76	523.92

NUTRIENT BUDGET

A. LAND APPLICATIONS

Bystrum 1 - 11/25/2022: Oats, silage-soft dough

Field name: Bystrum 1

Crop: Oats, silage-soft dough — Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	103.43	10.71	66.54	1,133.16	300,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		103.43	10.71	66.54	1,133.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.29	1.64	13.91	476.46	240,000.00 *gal*
Application event totals		93.29	1.64	13.91	476.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.29	1.64	13.91	476.46	240,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	24.67	2,160,000.00 *gal*
Application event totals		93.29	1.64	13.91	501.12	

Bystrum 1 - 06/20/2023: Corn, silage

Field name: Bystrum 1

Crop: Corn, silage — Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Bystrum 1 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	122.30	16.12	133.95	520.73	390,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	40.08	3,510,000.00 *gal*
Application event totals		122.30	16.12	133.95	560.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	33.92	2,970,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	30.83	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.77	12.72	97.24	1,048.83	300,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	30.83	2,700,000.00 *gal*
Application event totals		97.77	12.72	97.24	1,079.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	30.83	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	30.83	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.83	

Bystrum 1 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.77	12.72	97.24	1,048.83	300,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	30.83	2,700,000.00 *gal*
Application event totals		97.77	12.72	97.24	1,079.67	

09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	27.75	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.75	

Bystrum 2 - 11/25/2022: Oats, silage-soft dough

Field name: Bystrum 2

Crop: Oats, silage-soft dough

Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.30	11.63	72.24	1,230.29	600,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	0.00	5,400,000.00 *gal*
Application event totals		112.30	11.63	72.24	1,230.29	

01/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.30	1.12	9.44	323.31	300,000.00 *gal*
Application event totals		63.30	1.12	9.44	323.31	

Bystrum 2 - 11/25/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	113.95	2.01	17.00	581.96	540,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	30.13	4,860,000.00 *gal*
Application event totals		113.95	2.01	17.00	612.08	

Bystrum 2 - 06/20/2023: Corn, silage

Field name: Bystrum 2

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	122.57	16.15	134.24	521.87	720,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	40.17	6,480,000.00 *gal*
Application event totals		122.57	16.15	134.24	562.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	36.82	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	36.82	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.82	

Bystrum 2 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.46	14.50	110.85	1,195.67	630,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	35.15	5,670,000.00 *gal*
Application event totals		111.46	14.50	110.85	1,230.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	33.48	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	33.48	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.15	13.81	105.58	1,138.73	600,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	33.48	5,400,000.00 *gal*
Application event totals		106.15	13.81	105.58	1,172.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	30.13	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.13	

Bystrum 3 - 11/25/2022: Oats, silage-soft dough

Field name: Bystrum 3

Crop: Oats, silage-soft dough

Plant date: 11/25/2022

Bystrum 3 - 11/25/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.30	11.63	72.24	1,230.29	360,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	0.00	3,240,000.00 *gal*
Application event totals		112.30	11.63	72.24	1,230.29	

01/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.30	1.12	9.44	323.31	180,000.00 *gal*
Application event totals		63.30	1.12	9.44	323.31	

03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.50	1.86	15.74	538.85	300,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	27.90	2,700,000.00 *gal*
Application event totals		105.50	1.86	15.74	566.74	

Bystrum 3 - 06/20/2023: Corn, silage

Field name: Bystrum 3

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.42	16.27	135.18	525.50	435,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	40.45	3,915,000.00 *gal*
Application event totals		123.42	16.27	135.18	565.95	

Bystrum 3 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	39.05	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	36.26	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.99	14.96	114.37	1,233.63	390,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	36.26	3,510,000.00 *gal*
Application event totals		114.99	14.96	114.37	1,269.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	34.87	3,375,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	33.48	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.15	13.81	105.58	1,138.73	360,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	33.48	3,240,000.00 *gal*
Application event totals		106.15	13.81	105.58	1,172.21	

Bystrum 3 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	33.48	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.48	

Bystrum 4 - 11/25/2022: Oats, silage-soft dough

Field name: Bystrum 4

Crop: Oats, silage-soft dough

Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.98	12.63	78.47	1,336.35	540,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	0.00	4,860,000.00 *gal*
Application event totals		121.98	12.63	78.47	1,336.35	

01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.84	0.81	6.84	234.12	180,000.00 *gal*
Application event totals		45.84	0.81	6.84	234.12	

03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	122.24	2.15	18.23	624.32	480,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	32.32	4,320,000.00 *gal*
Application event totals		122.24	2.15	18.23	656.64	

Bystrum 4 - 06/20/2023: Corn, silage

Bystrum 4 - 06/20/2023: Corn, silage

Field name: Bystrum 4

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.28	16.25	135.02	524.87	600,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	40.40	5,400,000.00 *gal*
Application event totals		123.28	16.25	135.02	565.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	36.36	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	34.34	4,590,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	108.89	14.16	108.31	1,168.18	510,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	34.34	4,590,000.00 *gal*
Application event totals		108.89	14.16	108.31	1,202.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	33.33	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.33	

Bystrum 4 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	32.32	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	102.49	13.33	101.94	1,099.47	480,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	32.32	4,320,000.00 *gal*
Application event totals		102.49	13.33	101.94	1,131.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	32.32	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.32	

Bystrum 5 - 11/25/2022: Oats, silage-soft dough

Field name: Bystrum 5

Crop: Oats, silage-soft dough

Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.45	11.02	68.48	1,166.21	195,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	0.00	1,755,000.00 *gal*
Application event totals		106.45	11.02	68.48	1,166.21	

Bystrum 5 - 11/25/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.85	1.30	11.02	377.19	120,000.00 *gal*
Application event totals		73.85	1.30	11.02	377.19	

03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.78	1.95	16.52	565.79	180,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	29.29	1,620,000.00 *gal*
Application event totals		110.78	1.95	16.52	595.08	

Bystrum 5 - 06/20/2023: Corn, silage

Field name: Bystrum 5

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.17	15.71	130.52	507.38	240,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	39.05	2,160,000.00 *gal*
Application event totals		119.17	15.71	130.52	546.43	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	34.17	1,890,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.17	

Bystrum 5 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	34.17	1,890,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	108.36	14.09	107.78	1,162.46	210,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	34.17	1,890,000.00 *gal*
Application event totals		108.36	14.09	107.78	1,196.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	31.73	1,755,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	31.73	1,755,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.62	13.09	100.08	1,079.43	195,000.00 *gal*
TID Lateral 4	Surface water	0.00	0.00	0.00	31.73	1,755,000.00 *gal*
Application event totals		100.62	13.09	100.08	1,111.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Surface water	0.00	0.00	0.00	29.29	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Central 1 - 12/05/2022: Oats, silage-soft dough

Field name: Central 1

Crop: Oats, silage-soft dough

Plant date: 12/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/07/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	221.40	93.70	258.15	3,298.01	380.00 *ton*
Application event totals		221.40	93.70	258.15	3,298.01	

04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	305.95	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	305.95	

Central 1 - 07/18/2023: Corn, silage

Field name: Central 1

Crop: Corn, silage

Plant date: 07/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	214.37	143.47	417.05	3,653.38	420.00 *ton*
Application event totals		214.37	143.47	417.05	3,653.38	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	318.70	10,125,000.00 *gal*
Application event totals		0.00	0.00	0.00	318.70	

Central 1 - 07/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	305.95	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	305.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	305.95	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	305.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	305.95	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	305.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	293.21	9,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	293.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	280.46	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	280.46	

Central 2 - 12/05/2022: Oats, silage-soft dough

Field name: Central 2

Crop: Oats, silage-soft dough

Plant date: 12/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Central 2 - 12/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/07/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	246.70	104.41	287.65	3,674.93	260.00 *ton*
Application event totals		246.70	104.41	287.65	3,674.93	

04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	352.94	6,885,000.00 *gal*
Application event totals		0.00	0.00	0.00	352.94	

Central 2 - 07/18/2023: Corn, silage

Field name: Central 2

Crop: Corn, silage

Plant date: 07/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	216.11	144.64	420.46	3,683.20	260.00 *ton*
Application event totals		216.11	144.64	420.46	3,683.20	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	332.18	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	332.18	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

Central 2 - 07/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	280.28	5,467,500.00 *gal*
Application event totals		0.00	0.00	0.00	280.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	269.90	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	269.90	

Central 3 - 12/05/2022: Oats, silage-soft dough

Field name: Central 3

Crop: Oats, silage-soft dough

Plant date: 12/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	234.17	99.11	273.04	3,488.28	550.00 *ton*
Application event totals		234.17	99.11	273.04	3,488.28	

Central 3 - 12/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	335.37	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	335.37	

Central 3 - 07/18/2023: Corn, silage

Field name: Central 3

Crop: Corn, silage

Plant date: 07/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	216.33	144.78	420.87	3,686.84	580.00 *ton*
Application event totals		216.33	144.78	420.87	3,686.84	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	344.69	14,985,000.00 *gal*
Application event totals		0.00	0.00	0.00	344.69	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	335.37	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	335.37	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	335.37	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	335.37	

Central 3 - 07/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	326.06	14,175,000.00 *gal*
Application event totals		0.00	0.00	0.00	326.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	316.74	13,770,000.00 *gal*
Application event totals		0.00	0.00	0.00	316.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	298.11	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	298.11	

Central 4 - 12/05/2022: Oats, silage-soft dough

Field name: Central 4

Crop: Oats, silage-soft dough

Plant date: 12/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	216.59	91.67	252.53	3,226.32	450.00 *ton*
Application event totals		216.59	91.67	252.53	3,226.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	326.46	12,555,000.00 *gal*
Application event totals		0.00	0.00	0.00	326.46	

Central 4 - 07/18/2023: Corn, silage

Field name: Central 4

Crop: Corn, silage

Plant date: 07/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	198.17	132.62	385.54	3,377.30	470.00 *ton*
Application event totals		198.17	132.62	385.54	3,377.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	336.99	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	336.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	315.93	12,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	315.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	315.93	12,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	315.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	294.87	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	294.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	294.87	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	294.87	

Central 4 - 07/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	284.34	10,935,000.00 *gal*
Application event totals		0.00	0.00	0.00	284.34	

Central 5 - 12/05/2022: Oats, silage-soft dough

Field name: Central 5

Crop: Oats, silage-soft dough

Plant date: 12/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	215.86	91.36	251.69	3,215.56	130.00 *ton*
Application event totals		215.86	91.36	251.69	3,215.56	

05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	326.99	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	326.99	

Central 5 - 07/18/2023: Corn, silage

Field name: Central 5

Crop: Corn, silage

Plant date: 07/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Central 5 - 07/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	203.65	136.29	396.20	3,470.71	140.00 *ton*
Application event totals		203.65	136.29	396.20	3,470.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	363.32	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	363.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	308.82	3,442,500.00 *gal*
Application event totals		0.00	0.00	0.00	308.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	290.66	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	290.66	

Central 5 - 07/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	272.49	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	272.49	

Central 6 - 12/05/2022: Oats, silage-soft dough

Field name: Central 6

Crop: Oats, silage-soft dough

Plant date: 12/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	231.46	97.96	269.88	3,447.92	460.00 *ton*
Application event totals		231.46	97.96	269.88	3,447.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	297.26	10,935,000.00 *gal*
Application event totals		0.00	0.00	0.00	297.26	

Central 6 - 07/18/2023: Corn, silage

Field name: Central 6

Crop: Corn, silage

Plant date: 07/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Central 6 - 07/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	198.36	132.75	385.91	3,380.56	450.00 *ton*
Application event totals		198.36	132.75	385.91	3,380.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	330.29	12,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	330.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	313.78	11,542,500.00 *gal*
Application event totals		0.00	0.00	0.00	313.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	308.27	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	308.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	308.27	11,340,000.00 *gal*
Application event totals		0.00	0.00	0.00	308.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	330.29	12,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	330.29	

Central 6 - 07/18/2023: Corn, silage

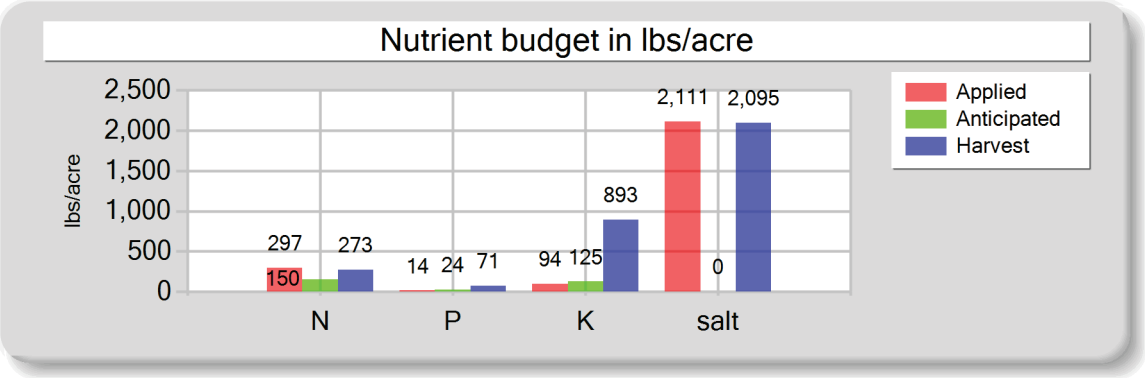
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 6	Surface water	0.00	0.00	0.00	291.76	10,732,500.00 *gal*
Application event totals		0.00	0.00	0.00	291.76	

B. NUTRIENT BUDGET

Bystrum 1 - 11/25/2022: Oats, silage-soft dough

Field name: Bystrum 1 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.01	14.00	94.37	2,086.07
Fresh water	0.00	0.00	0.00	24.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.01	14.00	94.37	2,110.74
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	272.81	70.88	892.76	2,094.80
Nutrient balance	24.20	-56.88	-798.39	15.94
Applied to removed ratio	1.09	0.20	0.11	1.01

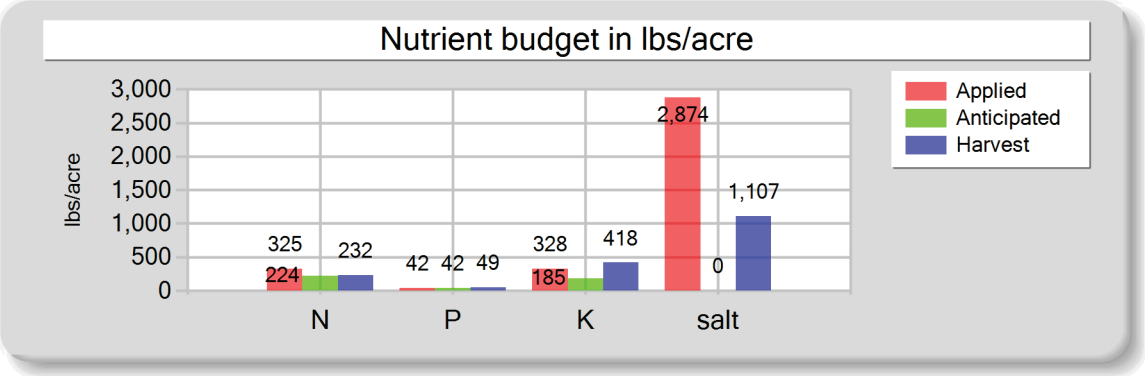
Fresh water applied	
4,860,000.00	*gallons*
178.98	*acre-inches*
9.42	*inches/acre*

Process wastewater applied	
780,000.00	*gallons*
28.72	*acre-inches*
1.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bystrum 1 - 06/20/2023: Corn, silage

Field name: Bystrum 1 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	317.84	41.55	328.43	2,618.40
Fresh water	0.00	0.00	0.00	255.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	324.84	41.55	328.43	2,874.31
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	232.47	49.33	418.31	1,106.94
Nutrient balance	92.37	-7.78	-89.88	1,767.36
Applied to removed ratio	1.40	0.84	0.79	2.60

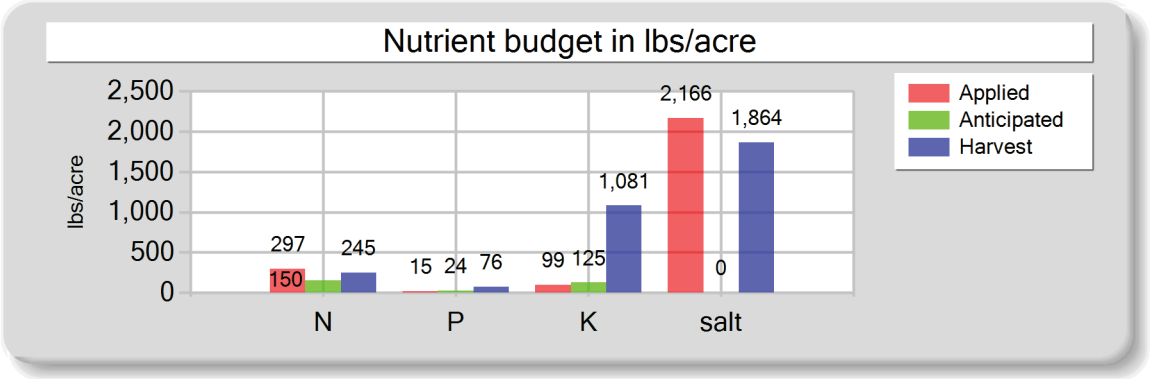
Fresh water applied	
22,410,000.00	*gallons*
825.28	*acre-inches*
43.44	*inches/acre*

Process wastewater applied	
990,000.00	*gallons*
36.46	*acre-inches*
1.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bystrum 2 - 11/25/2022: Oats, silage-soft dough

Field name: Bystrum 2 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	289.55	14.75	98.68	2,135.56
Fresh water	0.00	0.00	0.00	30.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.55	14.75	98.68	2,165.68
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	245.46	75.63	1,081.12	1,863.70
Nutrient balance	51.09	-60.88	-982.44	301.98
Applied to removed ratio	1.21	0.20	0.09	1.16

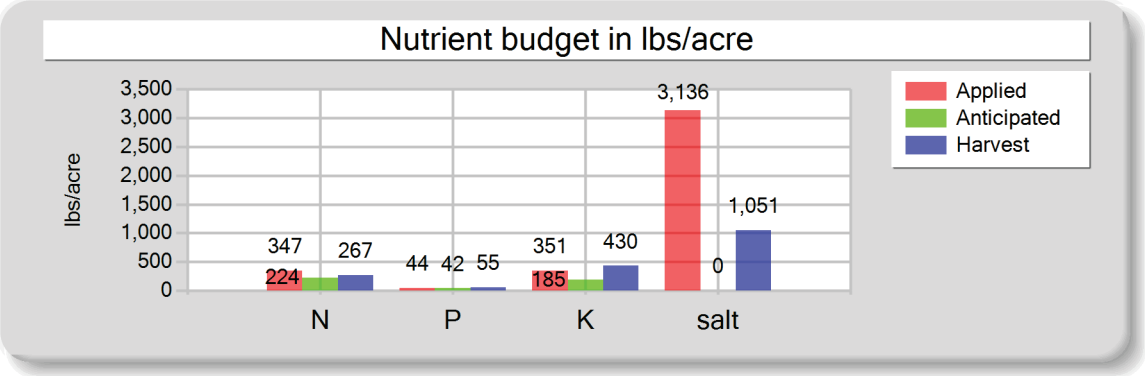
Fresh water applied	
10,260,000.00	*gallons*
377.84	*acre-inches*
10.80	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bystrum 2 - 06/20/2023: Corn, silage

Field name: Bystrum 2 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	340.18	44.45	350.68	2,856.28
Fresh water	0.00	0.00	0.00	279.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	347.18	44.45	350.68	3,135.80
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	267.32	54.91	430.24	1,051.41
Nutrient balance	79.86	-10.46	-79.57	2,084.38
Applied to removed ratio	1.30	0.81	0.82	2.98

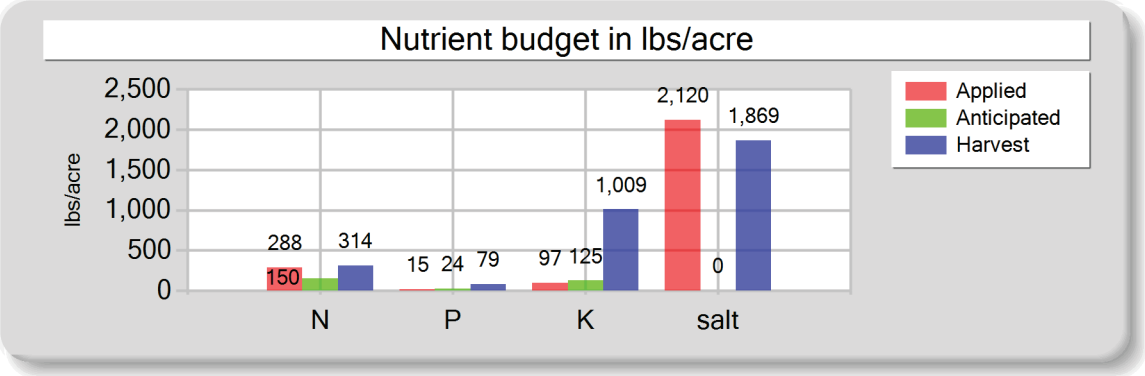
Fresh water applied	
45,090,000.00	*gallons*
1,660.51	*acre-inches*
47.44	*inches/acre*

Process wastewater applied	
1,950,000.00	*gallons*
71.81	*acre-inches*
2.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bystrum 3 - 11/25/2022: Oats, silage-soft dough

Field name: Bystrum 3 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	281.11	14.61	97.42	2,092.45
Fresh water	0.00	0.00	0.00	27.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.11	14.61	97.42	2,120.35
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	314.36	78.88	1,009.20	1,868.76
Nutrient balance	-26.25	-64.27	-911.78	251.59
Applied to removed ratio	0.92	0.19	0.10	1.13

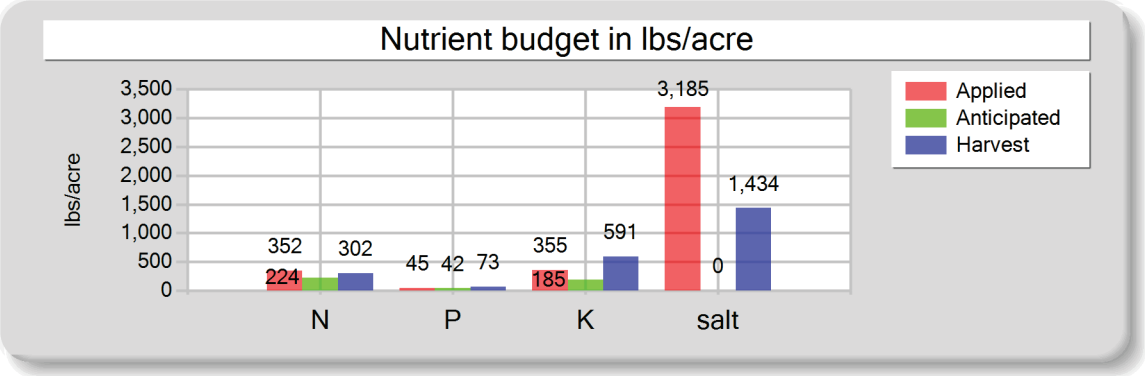
Fresh water applied	
5,940,000.00	*gallons*
218.75	*acre-inches*
10.42	*inches/acre*

Process wastewater applied	
840,000.00	*gallons*
30.93	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bystrum 3 - 06/20/2023: Corn, silage

Field name: Bystrum 3 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	344.57	45.03	355.13	2,897.86
Fresh water	0.00	0.00	0.00	287.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	351.57	45.03	355.13	3,185.19
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	301.96	72.98	590.52	1,434.12
Nutrient balance	49.61	-27.95	-235.39	1,751.07
Applied to removed ratio	1.16	0.62	0.60	2.22

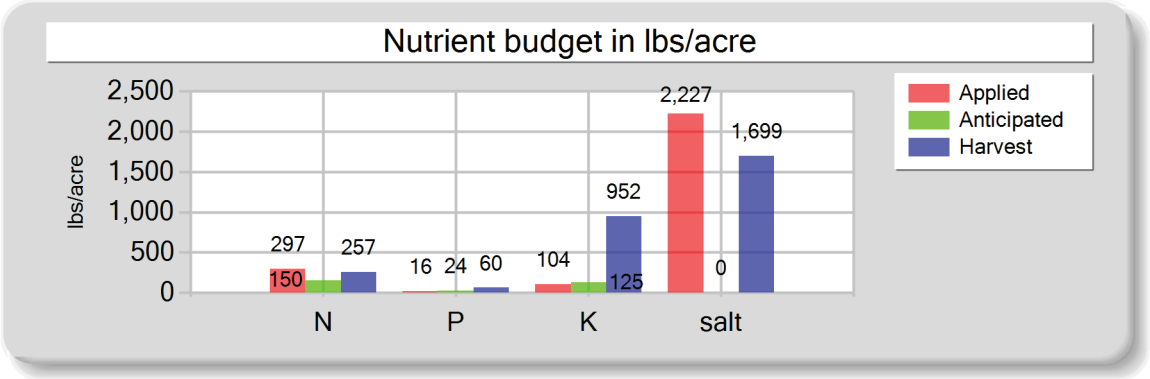
Fresh water applied	
27,810,000.00	*gallons*
1,024.15	*acre-inches*
48.77	*inches/acre*

Process wastewater applied	
1,185,000.00	*gallons*
43.64	*acre-inches*
2.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bystrum 4 - 11/25/2022: Oats, silage-soft dough

Field name: Bystrum 4 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.06	15.60	103.54	2,194.79
Fresh water	0.00	0.00	0.00	32.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.06	15.60	103.54	2,227.11
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	257.36	60.26	951.52	1,698.59
Nutrient balance	39.70	-44.67	-847.98	528.52
Applied to removed ratio	1.15	0.26	0.11	1.31

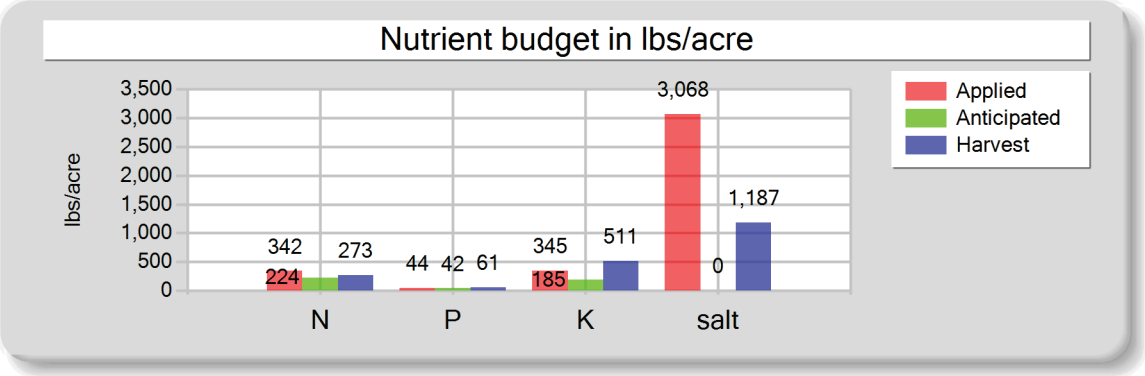
Fresh water applied	
9,180,000.00	*gallons*
338.07	*acre-inches*
11.66	*inches/acre*

Process wastewater applied	
1,200,000.00	*gallons*
44.19	*acre-inches*
1.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bystrum 4 - 06/20/2023: Corn, silage

Field name: Bystrum 4 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	334.66	43.74	345.26	2,792.52
Fresh water	0.00	0.00	0.00	275.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	341.66	43.74	345.26	3,068.26
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	272.51	60.56	511.03	1,187.27
Nutrient balance	69.15	-16.82	-165.77	1,881.00
Applied to removed ratio	1.25	0.72	0.68	2.58

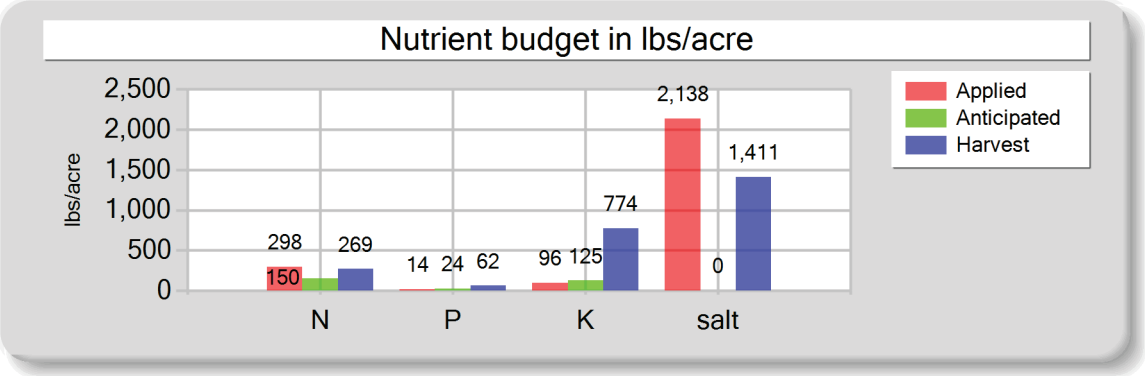
Fresh water applied	
36,855,000.00	*gallons*
1,357.24	*acre-inches*
46.80	*inches/acre*

Process wastewater applied	
1,590,000.00	*gallons*
58.55	*acre-inches*
2.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bystrum 5 - 11/25/2022: Oats, silage-soft dough

Field name: Bystrum 5 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	291.08	14.28	96.02	2,109.20
Fresh water	0.00	0.00	0.00	29.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.08	14.28	96.02	2,138.49
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	268.52	61.51	773.50	1,410.86
Nutrient balance	29.57	-47.23	-677.48	727.63
Applied to removed ratio	1.11	0.23	0.12	1.52

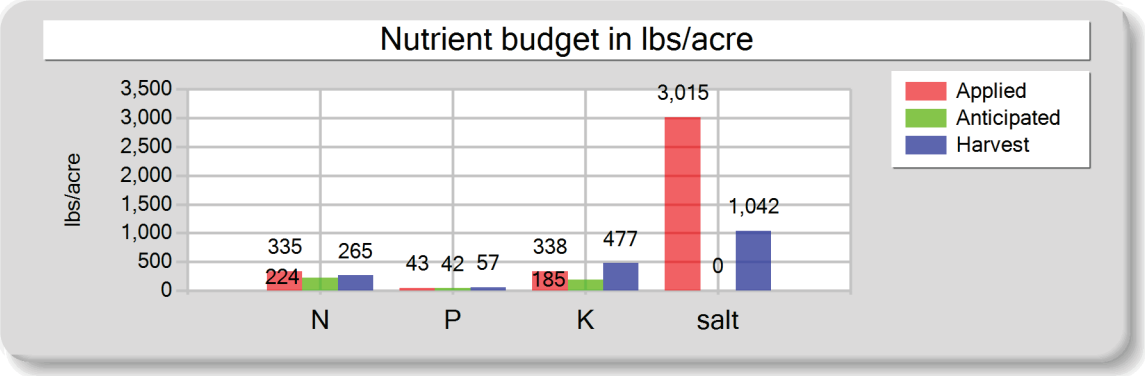
Fresh water applied	
3,375,000.00	*gallons*
124.29	*acre-inches*
10.36	*inches/acre*

Process wastewater applied	
495,000.00	*gallons*
18.23	*acre-inches*
1.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bystrum 5 - 06/20/2023: Corn, silage

Field name: Bystrum 5 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	328.15	42.88	338.37	2,749.26
Fresh water	0.00	0.00	0.00	266.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	335.15	42.88	338.37	3,015.32
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	265.00	56.74	476.60	1,041.70
Nutrient balance	70.15	-13.85	-138.23	1,973.62
Applied to removed ratio	1.26	0.76	0.71	2.89

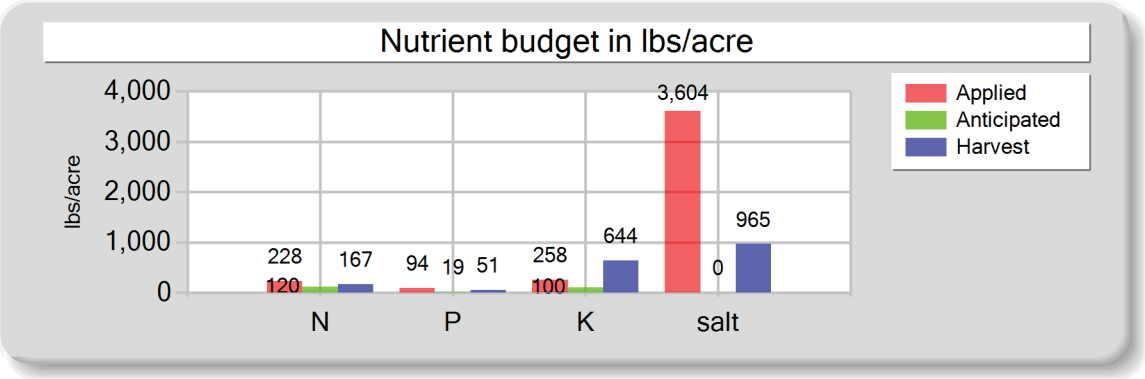
Fresh water applied	
14,715,000.00	*gallons*
541.90	*acre-inches*
45.16	*inches/acre*

Process wastewater applied	
645,000.00	*gallons*
23.75	*acre-inches*
1.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 1 - 12/05/2022: Oats, silage-soft dough

Field name: Central 1 Crop: Oats, silage-soft dough Plant date: 12/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	221.40	93.70	258.15	3,298.01
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	305.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.40	93.70	258.15	3,603.97
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	167.30	50.63	644.00	965.30
Nutrient balance	61.10	43.07	-385.85	2,638.67
Applied to removed ratio	1.37	1.85	0.40	3.73

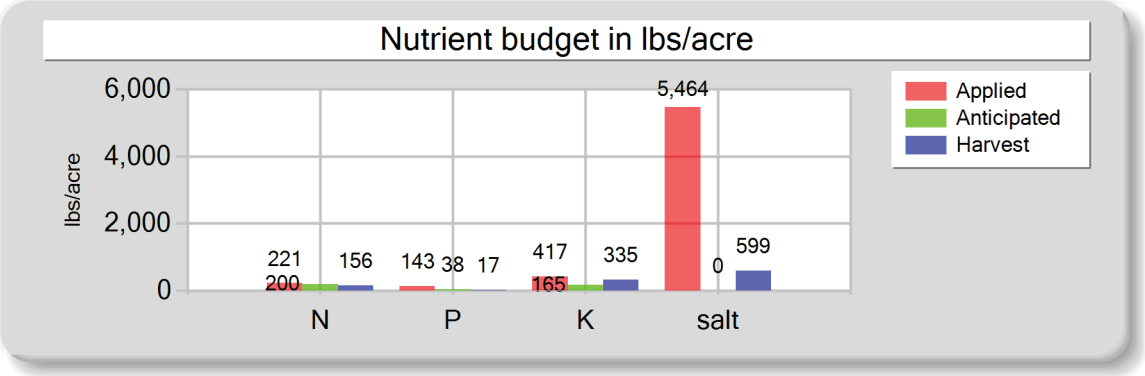
Fresh water applied	
9,720,000.00	*gallons*
357.95	*acre-inches*
6.28	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 1 - 07/18/2023: Corn, silage

Field name: Central 1 Crop: Corn, silage Plant date: 07/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	214.37	143.47	417.05	3,653.38
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,810.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.37	143.47	417.05	5,463.61
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	155.92	16.75	334.99	598.75
Nutrient balance	65.45	126.72	82.06	4,864.86
Applied to removed ratio	1.42	8.57	1.24	9.13

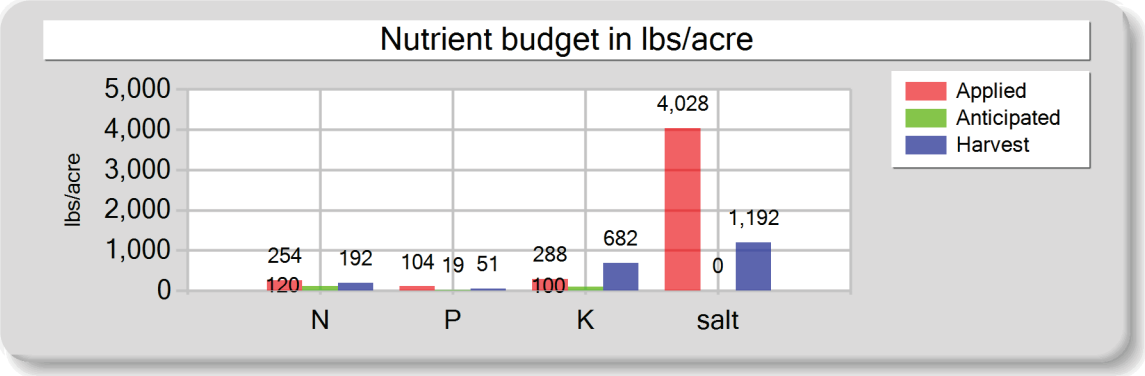
Fresh water applied	
57,510,000.00	*gallons*
2,117.90	*acre-inches*
37.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 2 - 12/05/2022: Oats, silage-soft dough

Field name: Central 2 Crop: Oats, silage-soft dough Plant date: 12/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	246.70	104.41	287.65	3,674.93
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	352.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	253.70	104.41	287.65	4,027.87
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	192.02	51.20	681.80	1,192.18
Nutrient balance	61.68	53.21	-394.15	2,835.69
Applied to removed ratio	1.32	2.04	0.42	3.38

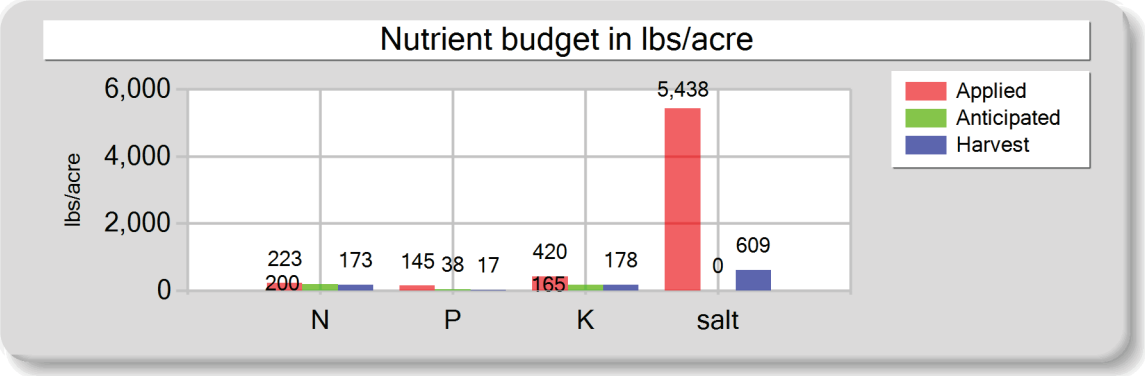
Fresh water applied	
6,885,000.00	*gallons*
253.55	*acre-inches*
7.24	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 2 - 07/18/2023: Corn, silage

Field name: Central 2 Crop: Corn, silage Plant date: 07/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	216.11	144.64	420.46	3,683.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,754.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.11	144.64	420.46	5,437.52
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	172.94	16.63	178.20	609.42
Nutrient balance	50.18	128.01	242.25	4,828.10
Applied to removed ratio	1.29	8.70	2.36	8.92

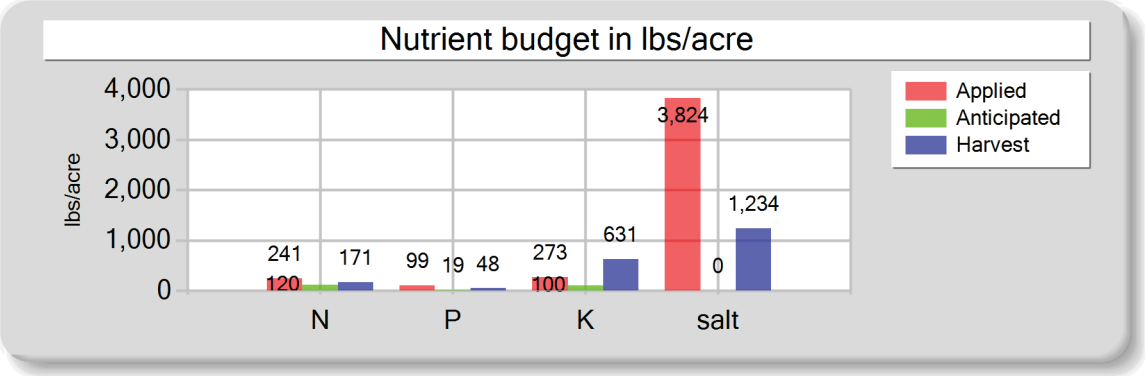
Fresh water applied	
34,222,500.00	*gallons*
1,260.30	*acre-inches*
36.01	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 3 - 12/05/2022: Oats, silage-soft dough

Field name: Central 3 Crop: Oats, silage-soft dough Plant date: 12/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	234.17	99.11	273.04	3,488.28
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	335.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.17	99.11	273.04	3,823.66
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	170.95	47.54	631.20	1,234.48
Nutrient balance	70.22	51.57	-358.16	2,589.17
Applied to removed ratio	1.41	2.08	0.43	3.10

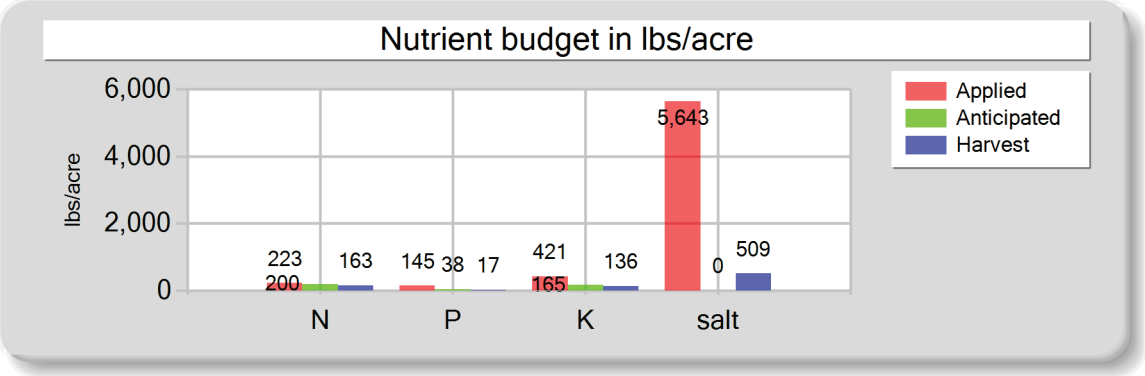
Fresh water applied	
14,580,000.00	*gallons*
536.93	*acre-inches*
6.88	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 3 - 07/18/2023: Corn, silage

Field name: Central 3 Crop: Corn, silage Plant date: 07/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	216.33	144.78	420.87	3,686.84
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,956.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.33	144.78	420.87	5,643.18
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	162.88	16.88	135.64	509.09
Nutrient balance	60.44	127.90	285.23	5,134.09
Applied to removed ratio	1.37	8.58	3.10	11.08

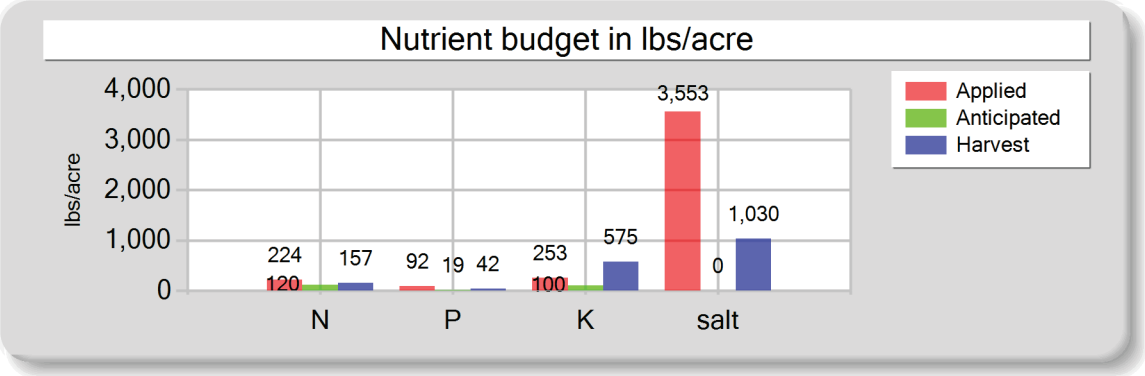
Fresh water applied	
85,050,000.00	*gallons*
3,132.10	*acre-inches*
40.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 4 - 12/05/2022: Oats, silage-soft dough

Field name: Central 4 Crop: Oats, silage-soft dough Plant date: 12/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	216.59	91.67	252.53	3,226.32
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	326.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.59	91.67	252.53	3,552.78
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	157.11	41.97	574.51	1,029.89
Nutrient balance	66.48	49.69	-321.97	2,522.89
Applied to removed ratio	1.42	2.18	0.44	3.45

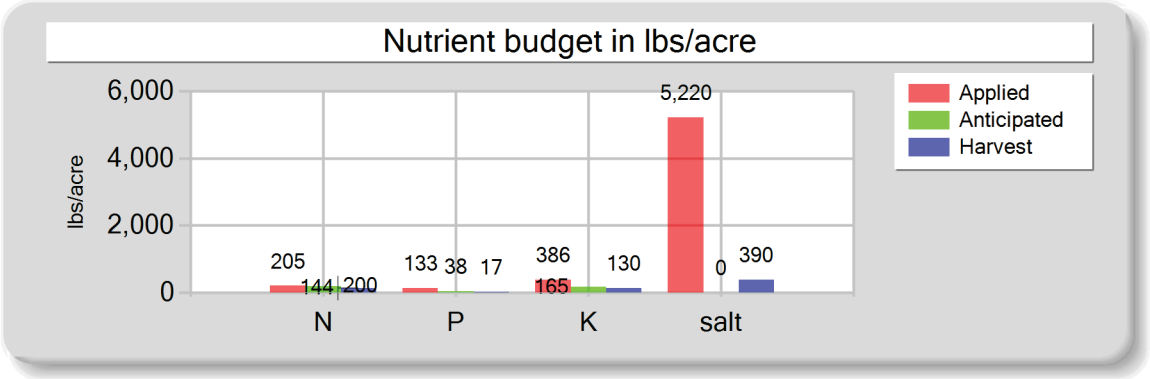
Fresh water applied	
12,555,000.00	*gallons*
462.36	*acre-inches*
6.70	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 4 - 07/18/2023: Corn, silage

Field name: Central 4 Crop: Corn, silage Plant date: 07/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	198.17	132.62	385.54	3,377.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,842.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.17	132.62	385.54	5,220.23
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	144.37	16.93	130.10	390.30
Nutrient balance	60.80	115.70	255.44	4,829.93
Applied to removed ratio	1.42	7.84	2.96	13.38

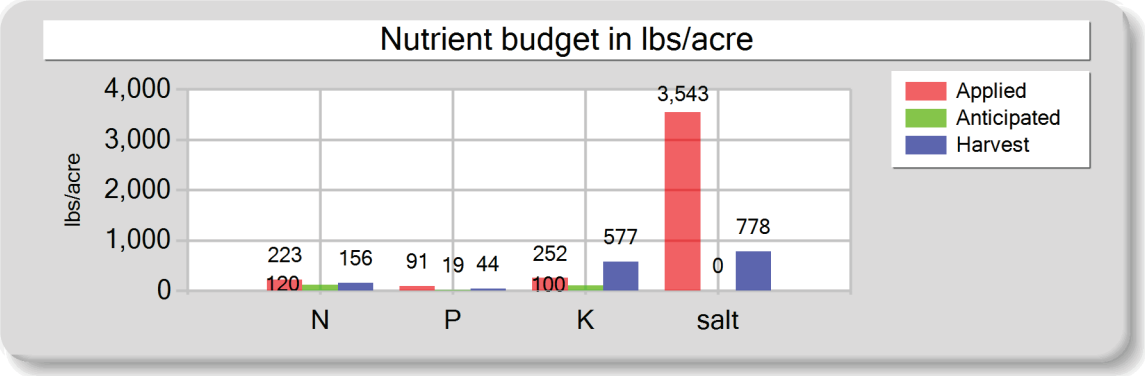
Fresh water applied	
70,875,000.00	*gallons*
2,610.09	*acre-inches*
37.83	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 5 - 12/05/2022: Oats, silage-soft dough

Field name: Central 5 Crop: Oats, silage-soft dough Plant date: 12/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	215.86	91.36	251.69	3,215.56
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	326.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.86	91.36	251.69	3,542.55
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	156.25	43.98	576.81	778.10
Nutrient balance	66.62	47.38	-325.12	2,764.45
Applied to removed ratio	1.43	2.08	0.44	4.55

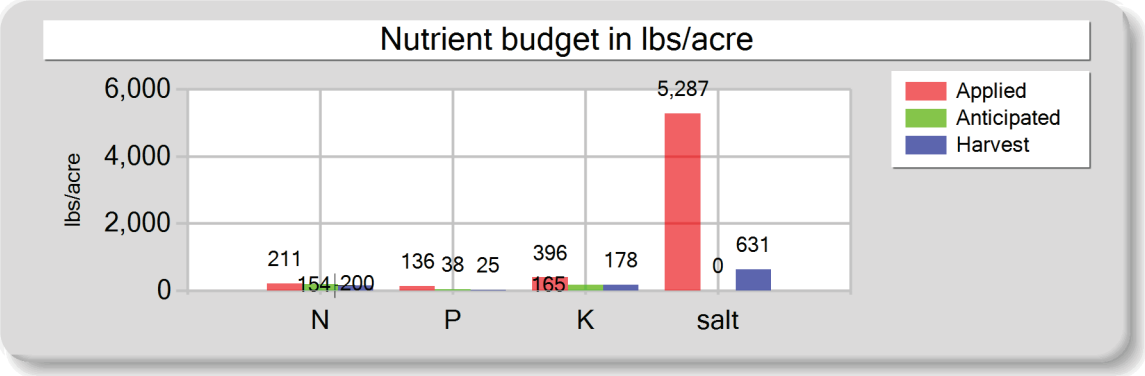
Fresh water applied	
3,645,000.00	*gallons*
134.23	*acre-inches*
6.71	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 5 - 07/18/2023: Corn, silage

Field name: Central 5 Crop: Corn, silage Plant date: 07/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	203.65	136.29	396.20	3,470.71
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,816.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.65	136.29	396.20	5,287.31
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	154.05	24.63	177.91	630.88
Nutrient balance	56.59	111.66	218.30	4,656.44
Applied to removed ratio	1.37	5.53	2.23	8.38

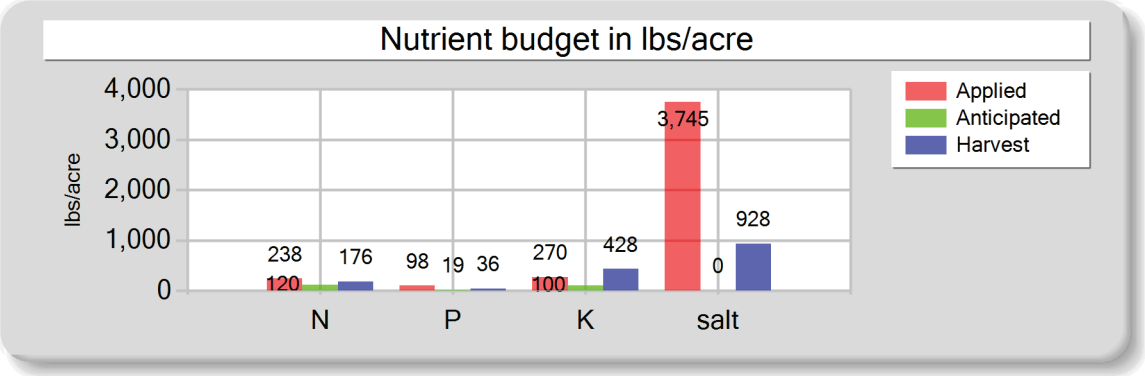
Fresh water applied	
20,250,000.00	*gallons*
745.74	*acre-inches*
37.29	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 6 - 12/05/2022: Oats, silage-soft dough

Field name: Central 6 Crop: Oats, silage-soft dough Plant date: 12/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	231.46	97.96	269.88	3,447.92
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	297.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.46	97.96	269.88	3,745.19
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	176.39	35.79	428.19	927.96
Nutrient balance	62.07	62.17	-158.31	2,817.23
Applied to removed ratio	1.35	2.74	0.63	4.04

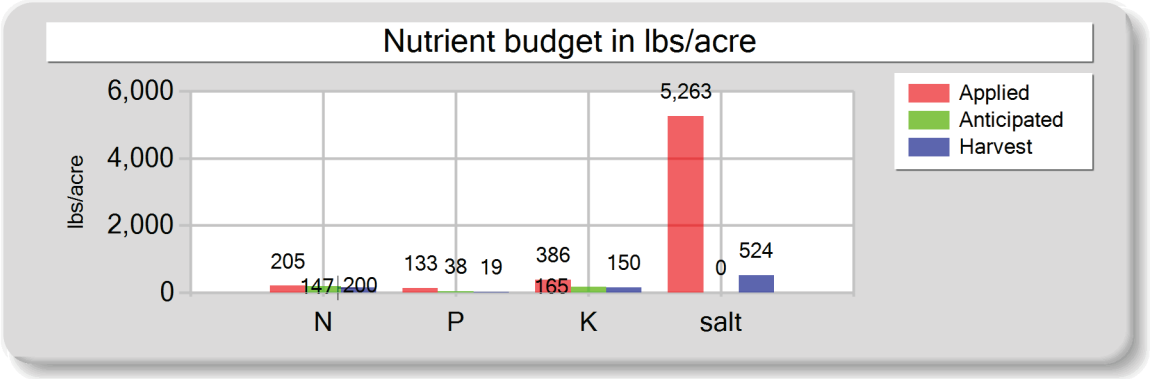
Fresh water applied	
10,935,000.00	*gallons*
402.70	*acre-inches*
6.10	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Central 6 - 07/18/2023: Corn, silage

Field name: Central 6 Crop: Corn, silage Plant date: 07/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	198.36	132.75	385.91	3,380.56
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,882.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.36	132.75	385.91	5,263.22
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	147.48	18.84	149.76	523.92
Nutrient balance	57.88	113.91	236.15	4,739.31
Applied to removed ratio	1.39	7.05	2.58	10.05

Fresh water applied	
69,255,000.00	*gallons*
2,550.43	*acre-inches*
38.64	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/13/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 31.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,100.00	10,200.00	28,100.00							35.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 43.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,700.00	17,200.00	50,000.00							43.80
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/27/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	785.00	440.00	0.00	0.00	81.30	505.00								7,470.00	8,600
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	885.00	319.00	0.00	0.00	15.60	132.00								6,390.00	4,520
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	714.00	445.00	0.00	0.00	94.10	782.00								7,710.00	3,040
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 06/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	742.00	249.00	0.00	0.00	96.50	738.00								5,370.00	7,960
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

TID Lateral 4

TID Lateral 4

Fresh Water

Sample description: Fresh Water

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								31.10	26
DL	0.50	0.50	0.50								1.00	10

TID Lateral 6

Fresh Water

Sample description: Fresh Water

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Bystrum 1 - 11/25/2022: Oats, silage-soft dough

Bystrum 1 - 11/25/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,120.00	1,850.00	23,300.00		13.60
DL	100.00	100.00	100.00		0.01

Bystrum 1 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/09/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,440.00	730.00	6,190.00		4.50
DL	100.00	100.00	100.00		0.01

Bystrum 2 - 11/25/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,290.00	1,630.00	23,300.00		13.30
DL	100.00	100.00	100.00		0.01

Bystrum 2 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/09/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,430.00	910.00	7,130.00		4.40
DL	100.00	100.00	100.00		0.01

Bystrum 3 - 11/25/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,130.00	2,040.00	26,100.00		13.50
DL	100.00	100.00	100.00		0.01

Bystrum 3 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/09/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,510.00	1,090.00	8,820.00		5.10
DL	100.00	100.00	100.00		0.01

Bystrum 4 - 11/25/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,680.00	1,330.00	21,000.00		13.20
DL	100.00	100.00	100.00		0.01

Bystrum 4 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/09/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,410.00	980.00	8,270.00		5.11
DL	100.00	100.00	100.00		0.01

Bystrum 5 - 11/25/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,290.00	1,670.00	21,000.00		13.30
DL	100.00	100.00	100.00		0.01

Bystrum 5 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/09/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,970.00	850.00	7,140.00		4.59
DL	100.00	100.00	100.00		0.01

Central 1 - 12/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,170.00	2,170.00	27,600.00		10.50
DL	100.00	100.00	100.00		0.01

Central 1 - 07/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,330.00	680.00	13,600.00		6.79
DL	100.00	100.00	100.00		0.01

Central 2 - 12/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,900.00	1,840.00	24,500.00		12.60
DL	100.00	100.00	100.00		0.01

Central 2 - 07/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,240.00	600.00	6,430.00		5.47
DL	100.00	100.00	100.00		0.01

Central 3 - 12/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,450.00	2,350.00	31,200.00		13.50
DL	100.00	100.00	100.00		0.01

Central 3 - 07/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	570.00	4,580.00		4.50
DL	100.00	100.00	100.00		0.01

Central 4 - 12/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,700.00	1,790.00	24,500.00		12.00
DL	100.00	100.00	100.00		0.01

Central 4 - 07/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,350.00	510.00	3,920.00		3.00
DL	100.00	100.00	100.00		0.01

Central 5 - 12/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,070.00	1,990.00	26,100.00		10.80
DL	100.00	100.00	100.00		0.01

Central 5 - 07/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,940.00	630.00	4,550.00		4.61
DL	100.00	100.00	100.00		0.01

Central 6 - 12/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,280.00	1,680.00	20,100.00		13.20
DL	100.00	100.00	100.00		0.01

Central 6 - 07/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,780.00	660.00	5,310.00		4.99
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2021 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,540.00	580.00	4,610.00		4.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

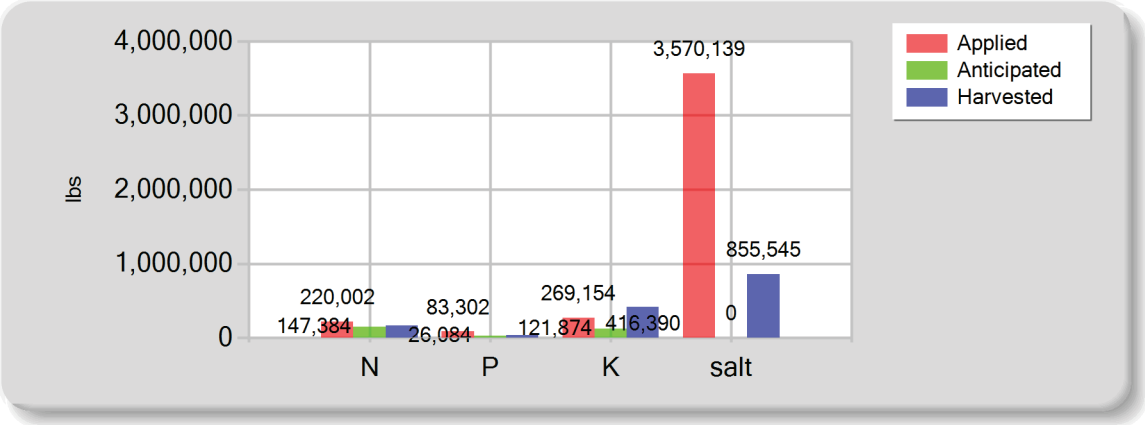
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

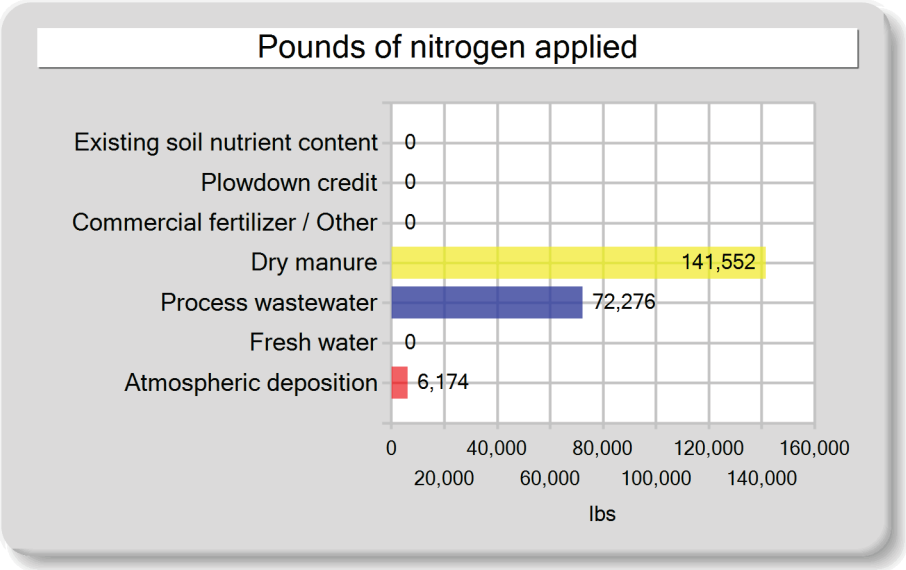
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

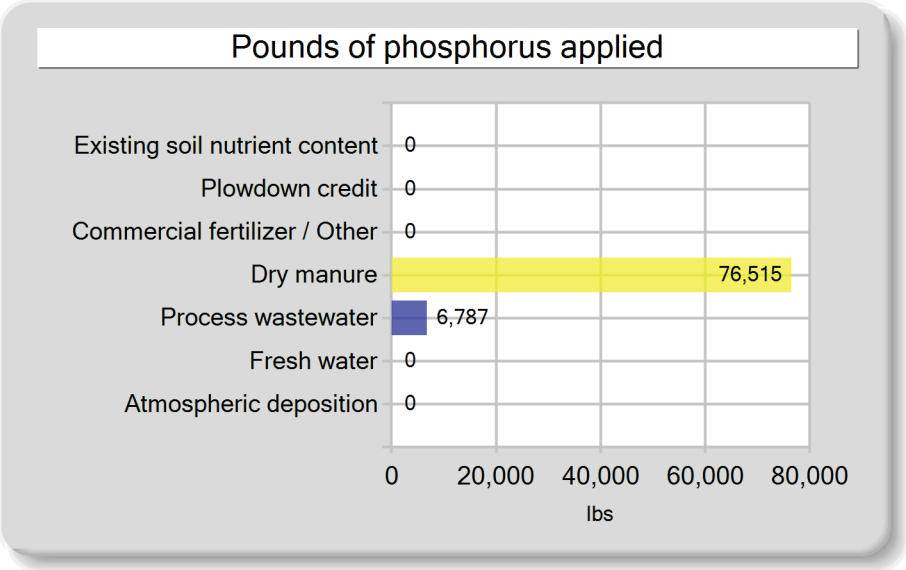
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141,552.22	76,515.32	217,661.61	2,253,478.58
Process wastewater	72,276.17	6,786.70	51,492.11	571,829.44
Fresh water	0.00	0.00	0.00	744,831.12
Atmospheric deposition	6,174.00	0.00	0.00	0.00
Total nutrients applied	220,002.39	83,302.02	269,153.73	3,570,139.14
Anticipated crop nutrient removal	147,384.00	26,083.50	121,873.80	0.00
Actual crop nutrient removal	167,877.93	35,246.89	416,389.66	855,544.87
Nutrient balance	52,124.46	48,055.13	-147,235.93	2,714,594.27
Applied to removed ratio	1.31	2.36	0.65	4.17

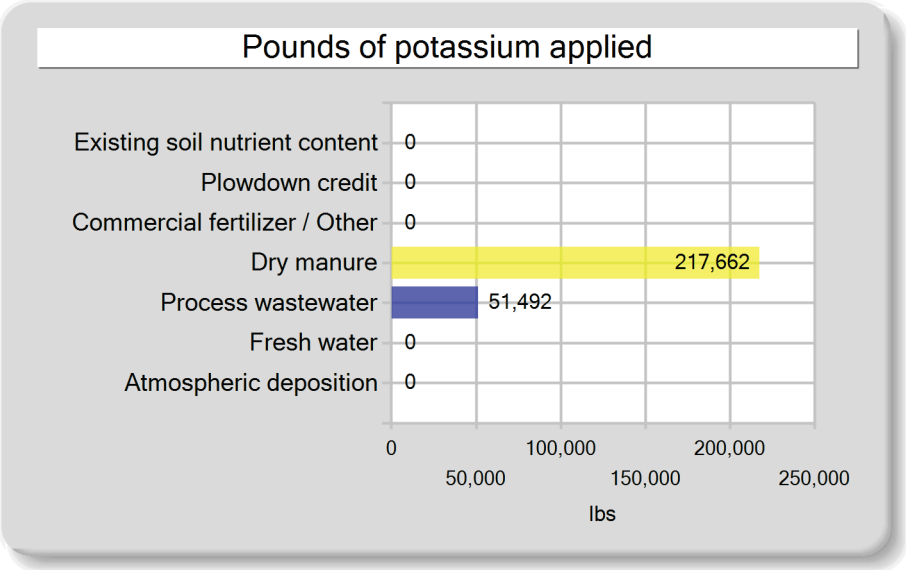
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

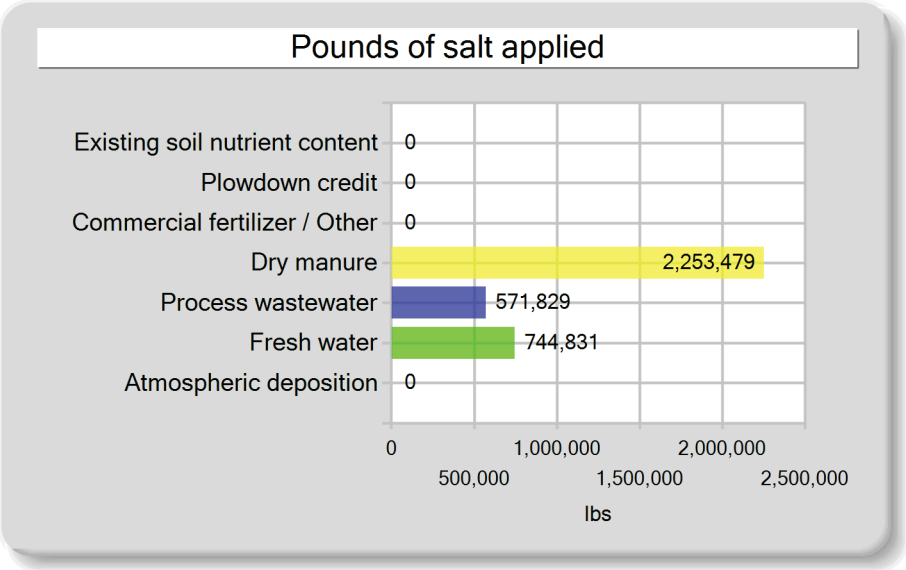


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Luis C Melo	Brad Nyman
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-19-24	6-19-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted.

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed.

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Live Oak Dairy G
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:33 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 1	T3240021A	Electrical Conductivity (EC)	EPA 120.1 and	1.11	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	712	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)	3.04
Electrical Conductivity (EC)	93.9			0.343
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)	2.36

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Live Oak Dairy G
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:33 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 10.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 2	T3240021B	Electrical Conductivity (EC)	EPA 120.1 and	1.12	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	23.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	714	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)	3.04
Electrical Conductivity (EC)	93.9			0.343
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)	2.36

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Compliance Analysis Report

Customer
Live Oak Dairy G
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:33 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 3	T3240021C	Electrical Conductivity (EC)	EPA 120.1 and	1.11	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	16.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	710	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)	3.04
Electrical Conductivity (EC)	93.9			0.343
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)	2.36

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Compliance Analysis Report

Customer
Live Oak Dairy G
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:33 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 4	T3240021D	Electrical Conductivity (EC)	EPA 120.1 and	1.11	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	20.9	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	712	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)	3.04
Electrical Conductivity (EC)	93.9			0.343
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)	2.36

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Live Oak Dairy G
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:33 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 8	T3240021E	Electrical Conductivity (EC)	EPA 120.1 and	1.10	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	705	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)	3.04
Electrical Conductivity (EC)	93.9			0.343
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)	2.36

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Compliance Analysis Report

Customer
Live Oak Dairy G
P.O. Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:33 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW R1	T3240021F	Electrical Conductivity (EC)	EPA 120.1 and	1.08	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	694	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	100 (T3240019A)	97.0 (T3240019A)	3.04
Electrical Conductivity (EC)	93.9			0.343
Nitrate Nitrogen (NO3-N)	98.8	95.3 (W3236004A)	93.1 (W3236004A)	2.36

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Live Oak G
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: BN
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/25	9:00	Domestic	DW 1	11.8°		✓				
2	8/25	9:05	wells	DW2	10.0°		✓				
3	8/25	9:10		DW 3	11.8°		✓				
4		9:15		DW 4	12.0°		✓				
5		9:20		DW 8	11.0°		✓				
6	✓	9:25	✓	DWR1	11.0°		✓				
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st		FRAG				
2nd		DENELE ANALYTICAL, INC.	8-28-23 @ 12:33			T3240021

Required to be submitted to any agency? (YES / NO)